Frobenius Lifting, Floquet Rigidity, and Equivariant Stabilization

Five theorems and their applications to prior open problems

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Abstract

We develop explicit constructions in three settings. For the lacunary series $\Phi_p(z) = \sum_{n\geq 0} z^{p^n}$, a Frobenius-compatible correction tower produces a monic polynomial basis with exact evaluation valuations $v_p(d!)$. We determine the full kernel of evaluation modulo every prime power and prove the minimum-degree conjecture of Krattenthaler and Müller, first posed in 2015. A multiplicative version, followed by a finite descent of denominators, proves their 2013 minimum-degree conjecture for the Cantor product $\prod_{n\geq 0}(1+z^{3^n})$ in its original coefficient ring. For discrete Schrödinger operators, coordinate-average descent and the second and third Fourier perturbation identities prove rigidity at period $(3\mathbb{Z})^d$ in every dimension. Combined with prior results, this completes the classification for rectangular period lattices. Finally, for an order-two lattice action, we construct mixed linear and quadratic systems of parameters with an optimal quadratic budget. Their Artinian quotients prove Clarke's coefficient inequality and give the sharp universal stabilization exponent $\max(p-q,0)$. These results comprise three numbered-conjecture applications and one explicitly posed, unnumbered classification problem. The five theorem labels do not denote five independently posed open problems.

1 Research questions, exact targets, and scope

The organizing strategy is to pose an auxiliary construction problem, solve it with explicit control of the quantity that obstructs a prior problem, and then return to that original problem without changing its hypotheses. The auxiliary quantities here are evaluation valuation, low-order Fourier constraints, and the number of quadratic parameters.

Label	Structural result	External application
T76	A factorial-valued monic basis, exact valuation formula, and the full evaluation kernel.	Auxiliary construction used in T77 and adapted in T78.
T77	Degree-optimal integral annihilators for every prime power.	Krattenthaler–Müller Conjecture 4, first preprint 2015 [1].
T78	Multiplicative lifting with finite coefficient-ring descent.	Krattenthaler–Müller Conjecture 12, 2013 [2].
T79	All-dimensional period-three rigidity and the rectangular classification.	The remaining small-period problem in [4, Section 4.4].
T80	Optimal mixed-degree parameter systems and sharp universal stabilization.	Clarke Conjecture 8.5, 2026 [6].

The two arithmetic targets concern formal series modulo integer powers, not the arithmetic nature of a numerical value. The spectral target is the rectangular-period problem, not

the larger classification over arbitrary oblique lattices. The Ehrhart result applies without assuming that the equivariant numerator is already polynomial or effective. These distinctions are retained throughout the proofs.

The claims below are unconditional mathematical statements with complete arguments, using the established results explicitly cited. In particular, the Cantor lower bound is the prior Theorem 10 of [2]; the remaining rectangular cases use the prior period-reduction and existence theorems of [4]; and the Ehrhart argument uses the Cohen–Macaulay property of normal affine semigroup rings. Exact finite computations supplement these arguments and are not used to extrapolate an all-parameter claim.

This is a research draft, independently checked internally but not externally peer reviewed or proof-assistant certified. The current literature search located no earlier identical resolution of the stated targets. The Frobenius congruence, Fourier perturbation, pole extraction, generic sections, and Koszul complexes are established mathematics. We therefore do not assert that the underlying general principles are historically unprecedented. The attribution and success-condition audit at the end makes this limitation explicit.

2 A Frobenius-compatible lifting problem

The first two existing targets concern sharp algebraic relations modulo prime powers for transcendental formal series. Krattenthaler and Müller [1, Conjecture 4] ask for the minimum degree of a monic polynomial annihilating

$$\Phi_p(z) = \sum_{n\geq 0} z^{p^n}$$

coefficientwise modulo p^γ, with coefficients in $\mathbb{Z}[z]$. Their proposed answer is the least d satisfying $\nu_p(d!) \geq \gamma$. Here p is an arbitrary prime and $\gamma \geq 1$. The binary version is also present in [3, Conjecture 4]; it is a special case of this target, not an additional independent problem. The second target, [2, Conjecture 12], concerns the Cantor product

$$\Psi(z) = \prod_{j\geq 0}(1+z^{3^j}).$$

Its prescribed coefficient ring is $\mathbb{Z}[z, z^{-1}, (1+z)^{-1}]$, and the proposed minimum degree modulo 3^γ is $2d$, for the least positive d with $d + \nu_3(d!) \geq \gamma$.

We begin with an auxiliary question that provides the same constructive step for both targets: given an integral series with an affine functional equation under $z \mapsto z^p$, can a monic relation of known divisibility be lifted recursively while controlling its degree? The answer uses the classical Frobenius congruence

$$B(z)^p \equiv B(z^p) \pmod{p} \qquad (B \in \mathbb{Z}[[z]]). \tag{1}$$

The resulting substitution-dependent recursion supplies the correction polynomials explicitly. The underlying congruence is classical; its use here is not described as a historically new foundational principle.

For an integer polynomial or integral formal series G, write $\nu_p(G)$ for the minimum p-adic valuation of its coefficients, with $\nu_p(0) = +\infty$. Congruence of formal series always means coefficientwise congruence.

Lemma 2.1 (Lifting through an affine Frobenius equation)**.** *Let $R \subseteq \mathbb{Z}[[z]]$ be a ring stable under the substitution $\sigma(c)(z) = c(z^p)$. Suppose $F \in \mathbb{Z}[[z]]$ satisfies*

$$F(z^p) = a(z)F(z) + b(z), \qquad a, b \in R.$$

For $G \in R[t]$, define

$$(\tau G)(z,t) = \sum_j \sigma(c_j)(z)(a(z)t + b(z))^j \quad \text{if } G = \sum_j c_j(z)t^j.$$

For a nonnegative integer e, if G is monic of degree $D > 0$ and $G(z, F(z)) \in p^e\mathbb{Z}[[z]]$, then

$$\mathcal{L}_e(G) = G^p - p^{(p-1)e}\tau G \tag{2}$$

is monic of degree pD and has evaluation divisible by p^{pe+1}.

Starting with a monic G_0 of degree $D_0 > 0$ and evaluation divisible by p^{e_0} for a nonnegative integer e_0, set

$$G_{r+1} = G_r^p - p^{(p-1)e_r}\tau G_r, \qquad e_{r+1} = pe_r + 1. \tag{3}$$

Then G_r is monic of degree D_0p^r and has evaluation divisible by p^{e_r}, where

$$e_r = p^r e_0 + \frac{p^r - 1}{p - 1}.$$

If $d = \sum_r d_r p^r$ is a base-p expansion, the product $\prod_r G_r^{d_r}$ is monic of degree $D_0 d$ and has evaluation divisible by $p^{e_0 d + \nu_p(d!)}$.

Proof. Write $G(z, F(z)) = p^e B(z)$ with $B \in \mathbb{Z}[[z]]$. The affine functional equation makes evaluation commute with the two substitutions:

$$(\tau G)(z, F(z)) = G(z^p, F(z^p)) = p^e B(z^p).$$

Consequently

$$\mathcal{L}_e(G)(z, F(z)) = p^{pe}\big(B(z)^p - B(z^p)\big),$$

which is divisible by p^{pe+1} by (1). The correction τG has t-degree at most D, whereas G^p is monic of degree $pD > D$. This proves the first assertion and permits induction in (3). The recurrence for e_r has the displayed solution. Finally, Legendre's formula gives

$$\sum_r d_r e_r = e_0 d + \sum_r d_r \frac{p^r - 1}{p - 1} = e_0 d + \frac{d - s_p(d)}{p - 1} = e_0 d + \nu_p(d!),$$

where $s_p(d) = \sum_r d_r$. Multiplying the evaluation divisibilities proves the last assertion. □

3 An exact factorial basis and its evaluation kernel

For Φ_p, the exact identity

$$\Phi_p(z^p) = \Phi_p(z) - z \tag{4}$$

allows us to take $R = \mathbb{Z}[z]$, $a = 1$, $b = -z$, and the seed $A_0 = t$. Put $e_r = (p^r - 1)/(p - 1)$ and define

$$A_0(z,t) = t, \qquad A_{r+1}(z,t) = A_r(z,t)^p - p^{(p-1)e_r} A_r(z^p, t - z). \tag{5}$$

All substitutions in this formula are simultaneous. For $d \geq 0$ with base-p digits d_r, set

$$B_d(z,t) = \prod_{r \geq 0} A_r(z,t)^{d_r}; \qquad B_0 = 1. \tag{6}$$

Only finitely many factors occur. Lemma 2.1 implies that B_d is monic of degree d and that

$$\nu_p\big(B_d(z, \Phi_p(z))\big) \geq \nu_p(d!). \tag{7}$$

The opposite bound will follow from an explicit coefficient-isolation argument. In particular, it does not require a linear-independence theorem for a separate family of auxiliary series.

Lemma 3.1 (Isolation by digit sums)**.** *Let $H(z,t)=\sum_{k=0}^{D} c_k(z)t^k \in \mathbb{Z}[z,t]$, where $D \geq 1$, and choose $M \geq 0$ with $\deg c_k \leq M$ for every nonzero c_k. Choose $L \geq 0$ such that $p^L > M$, set $K = L+2$, and define*

$$N = \sum_{i=0}^{D-1} p^{K+i}.$$

Then, for every $j_0 \in \{0,\dots,M\}$,

$$[z^{N+j_0}]H(z,\Phi_p(z)) = D!\,[z^{j_0}]c_D(z). \tag{8}$$

In particular, if $H(z,\Phi_p(z)) \equiv 0 \pmod{p^\gamma}$, then every coefficient of $D!c_D(z)$ is divisible by p^γ.

Proof. Every exponent appearing in $\Phi_p(z)^k$ is a sum of k powers of p. Its base-p digit sum is at most k, since every carry decreases the number of summands by $p-1$.

Consider the term $[z^j]c_k(z)$ in a coefficient extraction at $N+j_0$, and put $\Delta = j_0 - j$. If $\Delta > 0$, then $\Delta \leq M < p^K$ and

$$s_p(N+\Delta) = D + s_p(\Delta) > D.$$

If $\Delta = -c < 0$, then $1 \leq c \leq M < p^L$ and borrowing at the lowest nonzero digit of N gives

$$\begin{aligned} s_p(N-c) &= D - 1 + s_p(p^K - c) \\ &= D - 1 + (p-1)K - s_p(c-1) \\ &\geq D - 1 + (p-1)(K-L) > D. \end{aligned}$$

Here $s_p(c-1) \leq (p-1)L$, and $K - L = 2$. Thus, whenever $j \neq j_0$, the required coefficient in every Φ_p^k, $k \leq D$, is zero.

For $j = j_0$, the exponent is N and has digit sum D. Therefore only $k = D$ can contribute. A representation of N by exactly D powers of p cannot involve a carry: a carry would lower the final digit sum below D. Its multiset of summands is therefore precisely the D distinct powers in the definition of N. Their ordered arrangements number $D!$. This proves (8) and the divisibility assertion. □

For later statements, define

$$D_\gamma = \min\{d \geq 0 : \nu_p(d!) \geq \gamma\}, \qquad J_\gamma = \{H \in \mathbb{Z}[z,t] : H(z,\Phi_p(z)) \equiv 0 \pmod{p^\gamma}\}.$$

The set J_γ is an ideal because evaluation modulo p^γ is a ring homomorphism.

Theorem T76 (Factorial basis, exact valuation, and the full kernel)**.** *The polynomials B_d in (6) form a monic triangular $\mathbb{Z}[z]$-basis of $\mathbb{Z}[z,t]$. For every finite expansion $H = \sum_d c_d(z)B_d(z,t)$,*

$$\nu_p(H(z,\Phi_p(z))) = \min_d(\nu_p(c_d) + \nu_p(d!)). \tag{9}$$

In particular, each B_d has evaluation valuation exactly $\nu_p(d!)$. For every $\gamma \geq 1$, membership in J_γ is equivalent to

$$c_d(z) \in p^{\max\{0,\gamma-\nu_p(d!)\}}\mathbb{Z}[z] \quad \text{for every } d. \tag{10}$$

The ideal has the finite presentation

$$J_\gamma = \left(B_{D_\gamma},\ p^{\gamma-\nu_p(d!)}B_d \ :\ 0 \leq d < D_\gamma\right), \tag{11}$$

and, as a $\mathbb{Z}[z]$-module,

$$\mathbb{Z}[z,t]/J_\gamma \simeq \bigoplus_{d=0}^{D_\gamma-1} (\mathbb{Z}/p^{\gamma-\nu_p(d!)}\mathbb{Z})[z]. \tag{12}$$

Moreover, giving t weight 1 and z weight p yields $\mathrm{wt}_p(B_d) \leq d$; consequently $\deg_z B_d \leq \lfloor d/p \rfloor$.

Proof. Since B_d is monic of t-degree d, successively subtracting its leading term gives a unique finite expansion of every polynomial in the B_d. This proves the basis assertion.

Sufficiency of (10) follows from (7). For necessity, suppose $H \in J_\gamma$ and let D be the largest index with $c_D \neq 0$. This is also its ordinary t-degree, with leading coefficient $c_D(z)$. If $D \geq 1$, Lemma 3.1 shows

$$c_D(z) \in p^{\max\{0,\gamma-\nu_p(D!)\}}\mathbb{Z}[z].$$

The same conclusion is immediate for $D = 0$. By (7), the evaluation of $c_D B_D$ vanishes modulo p^γ. Subtracting it leaves a smaller-degree element of J_γ. Induction proves (10) for every coefficient.

For $H \neq 0$, put $a = \min_d(\nu_p(c_d) + \nu_p(d!))$, a finite nonnegative integer. Equation (7) gives valuation at least a. If the valuation were at least $a+1$, criterion (10) with $\gamma = a+1$ would contradict the definition of a. This proves (9); the zero case uses the convention $\nu_p(0) = +\infty$.

Each displayed generator in (11) belongs to J_γ. Conversely, divide any $H \in J_\gamma$ by the monic B_{D_γ} in $\mathbb{Z}[z][t]$. The remainder is in J_γ and has degree below D_γ. Its unique basis expansion satisfies (10), placing it in the displayed ideal. Uniqueness of the remainder coefficients modulo their indicated powers of p also proves the module decomposition (12). This is a module isomorphism; no diagonal description of multiplication in the quotient ring is asserted.

Finally, $A_0 = t$ has weight one. The substitution $z \mapsto z^p$, $t \mapsto t - z$ multiplies the weight of every monomial by at most p: the substituted z has weight p^2 and the substituted t has weight at most p. Thus (5) gives $\mathrm{wt}_p(A_r) \leq p^r$ inductively. Multiplying according to the base-p digits of d yields $\mathrm{wt}_p(B_d) \leq \sum_r d_r p^r = d$, proving the last assertion. □

The coefficient-isolation identity explains why the upper bound from Frobenius lifting is sharp: it produces an exact coefficient $d!$ at a specified exponent for every monic polynomial of degree d once its coefficient degrees have been bounded. It also prevents cancellation between different valuation levels in the basis expansion.

4 The original minimum-degree conjecture for Φ_p

Theorem T77 (Krattenthaler–Müller Conjecture 4)**.** *For every prime p and every integer $\gamma \geq 1$, the smallest t-degree of a monic polynomial $H(z,t) \in \mathbb{Z}[z,t]$ satisfying*

$$H(z, \Phi_p(z)) \equiv 0 \pmod{p^\gamma}$$

is precisely

$$D_\gamma = \min\{d \geq 0 : \nu_p(d!) \geq \gamma\}.$$

A degree-optimal polynomial is B_{D_γ}, defined explicitly by (5)–(6); it has z-degree at most $\lfloor D_\gamma/p \rfloor$. This proves the full conjecture in [1], with its original coefficient ring and quantifiers.

Proof. Equation (7) shows that B_{D_γ} is a monic annihilator modulo p^γ of degree D_γ. Conversely, for a monic annihilator of degree $D \geq 1$, Lemma 3.1 gives $p^\gamma \mid D!$. Hence $D \geq D_\gamma$. Degree zero cannot occur: the only monic constant polynomial is one. The coefficient-polynomial degree bound follows from Theorem T76. □

For illustration, when $p = 2$, the recursion begins with

$$A_1(z,t) = t^2 - t + z,$$
$$A_2(z,t) = t^4 - 2t^3 + (2z-1)t^2 + (2z+2)t - 3z^2 - 2z.$$

The evaluations of these polynomials have valuations exactly one and three. Thus A_2 is a minimum-degree relation modulo 8. At the next levels, A_3 and A_4 have degrees eight and sixteen and evaluation valuations seven and fifteen. These statements hold for the full formal series, not only a truncation. The recurrence supplies the polynomials without solving an undetermined coefficient system at each level.

5 The original Cantor-product minimum-degree conjecture

For this second application, the affine functional equation is

$$\Psi(z^3) = \frac{\Psi(z)}{1+z}. \tag{13}$$

The original coefficient ring $R = \mathbb{Z}[z, z^{-1}, (1+z)^{-1}]$ is not stable under substitution $z \mapsto z^3$ as an ordinary rational-function ring. We first perform the construction in the stable ring

$$R_\infty = \mathbb{Z}\big[z, (1+z^{3^j})^{-1} : j \geq 0\big] \subseteq \mathbb{Z}[[z]],$$

and then descend every coefficient at the required finite precision. The following lemma ensures that the final answer uses the prescribed ring, rather than a changed problem.

Lemma 5.1 (Finite-precision descent)**.** *For every $\gamma \geq 1$ and every $c \in R_\infty$, there exists $\widetilde{c} \in \mathbb{Z}[z, (1+z)^{-1}]$ such that $c - \widetilde{c} \in 3^\gamma \mathbb{Z}[[z]]$. Consequently a monic polynomial in $R_\infty[t]$ can be replaced, without changing its degree or any evaluation modulo 3^γ at an integral formal series, by a monic polynomial in $\mathbb{Z}[z, (1+z)^{-1}][t]$.*

Proof. For $j \geq 0$ put $U_j = 1 + z^{3^j}$ and $V_j = (1+z)^{3^j}$. Frobenius modulo three gives $U_j = V_j + 3E_j$ for some $E_j \in \mathbb{Z}[z]$. The finite geometric identity gives

$$U_j^{-1} \equiv \sum_{a=0}^{\gamma-1} (-3E_j)^a V_j^{-a-1} \pmod{3^\gamma \mathbb{Z}[[z]]}. \tag{14}$$

All denominators on the right are powers of $1+z$. More explicitly, the omitted remainder is $(-3E_j)^\gamma V_j^{-\gamma} U_j^{-1}$, whose formal coefficients are divisible by 3^γ because both U_j and V_j have constant term one. Every $c \in R_\infty$ is a polynomial expression in z and finitely many U_j^{-1}. Substituting (14) in that expression proves the first assertion. For a polynomial in t, perform the replacement on its finitely many non-leading coefficients and keep the leading coefficient one unchanged. Evaluation at an integral series preserves the congruence. □

Set

$$\begin{aligned} C_0(z,t) &= t^2 - \frac{1}{1+z}, \qquad f_0 = 1, \\ C_{r+1}(z,t) &= C_r(z,t)^3 - 3^{2f_r} C_r\left(z^3, \frac{t}{1+z}\right), \qquad f_{r+1} = 3f_r + 1. \end{aligned} \tag{15}$$

The seed has evaluation divisible by three. Indeed, $\Psi(z)^3 \equiv \Psi(z^3) \pmod 3$ by Frobenius, and (13) implies $\Psi(z)^2 \equiv (1+z)^{-1} \pmod 3$ because Ψ has constant term one and is a unit of $\mathbb{Z}[[z]]$. Applying Lemma 2.1 inside R_∞ gives

$$\deg_t C_r = 2 \cdot 3^r, \qquad f_r = \frac{3^{r+1}-1}{2}, \qquad C_r(z, \Psi(z)) \in 3^{f_r}\mathbb{Z}[[z]], \tag{16}$$

with every C_r monic.

Theorem T78 (Krattenthaler–Müller Conjecture 12)**.** *Let $\gamma \geq 1$ and let*

$$d_\gamma = \min\{d \geq 1 : d + \nu_3(d!) \geq \gamma\}.$$

The smallest degree of a monic polynomial $H(z,t) \in \mathbb{Z}[z, z^{-1}, (1+z)^{-1}][t]$ satisfying

$$H(z, \Psi(z)) \equiv 0 \pmod{3^\gamma}$$

is exactly $2d_\gamma$. A polynomial attaining this degree is obtained from the base-three digit product of (15) followed by the finite descent in Lemma 5.1. In particular, the unchanged Conjecture 12 of [2] holds for every γ.

Proof. Write $d = d_\gamma = \sum_r d_r 3^r$, where $0 \leq d_r < 3$, and put $E_d = \prod_r C_r^{d_r} \in R_\infty[t]$. It is monic of degree $2d$. Using (16) and Legendre's formula,

$$\nu_3(E_d(z, \Psi(z))) \geq \sum_r d_r f_r = \frac{3d - s_3(d)}{2} = d + \nu_3(d!) \geq \gamma.$$

Lemma 5.1 replaces E_d by a monic polynomial $\widetilde{E}_d$ of the same degree in $\mathbb{Z}[z, (1+z)^{-1}][t]$, with the same evaluation modulo 3^γ. This coefficient ring is a subring of the ring prescribed in the original problem, so it supplies the required upper bound without enlarging the allowed ring.

For the reverse inequality, Theorem 10 of [2] proves that the minimum degree in precisely that prescribed ring is even, say $2b$, and must satisfy $b + \nu_3(b!) \geq \gamma$. Hence $b \geq d_\gamma$. Combined with the constructed degree-$2d_\gamma$ polynomial, this proves equality. The cited lower bound is an established theorem; the conjectural step in that source was attainment of the bound for all γ, which the construction above provides. □

An explicit example in the original ring is the degree-six polynomial

$$t^6 - \frac{3t^4}{1+z} - \frac{6t^2}{(1+z)^2} + \frac{8}{(1+z)^3} + \frac{27z}{(1+z)^5},$$

whose evaluation at Ψ vanishes modulo 81. To obtain it from C_1, use

$$\frac{1}{1+z^3} \equiv \frac{1}{(1+z)^3} + \frac{3z}{(1+z)^5} \pmod{9}.$$

Only precision nine is needed for this coefficient because it is multiplied by nine in C_1. Since $3 + \nu_3(3!) = 4$, the theorem certifies that degree six is minimal for modulus 3^4.

The two applications share the seed-lifting mechanism, but their sharpness arguments and coefficient rings differ. For Φ_p the coefficient isolation also determines the entire evaluation kernel and its module structure. For Ψ, finite-precision descent is essential to preserve the original problem's ring. The binary specialization of the first application remains nested within the all-prime result.

6 Small-period Floquet rigidity in every dimension

Let Δ be the discrete Laplacian on $\mathbb{Z}^d$ with nearest-neighbor coefficients equal to one. Two complex periodic potentials are called *Floquet isospectral* when their Floquet matrices have identical characteristic polynomials at every quasimomentum. We address the remaining rectangular-period problem explicitly identified in [4, Section 4.4]: whether a nonzero complex potential of period $(3\mathbb{Z})^d$, $d \geq 3$, can be Floquet isospectral to zero. That source singles out

$d = 3$ as the smallest unresolved case. This is an explicitly posed but unnumbered question. It is narrower than the classification over arbitrary oblique lattices, which is not settled here.

Our auxiliary question is whether coordinate averaging, followed by just the second and third perturbation identities, can force the potential to vanish. The proof below gives an affirmative answer. Fourier representations, rational spectral identities, and extraction of coefficients through their pole hyperplanes have prior applications to Floquet rigidity; see, in particular, [5, Lemma 3.2, Theorem 3.3, and Section 4]. The contribution proposed here is their combination with coordinate-average descent and the specific third-order constraint for the period-three Fourier support.

Theorem T79 (Period-three rigidity and the rectangular classification)**.** *For every $d \geq 1$, a complex $(3\mathbb{Z})^d$-periodic potential V on $\mathbb{Z}^d$ that is Floquet isospectral to zero satisfies $V \equiv 0$. Consequently, for arbitrary positive integers $q_1, \ldots, q_d$, there exists a nonzero complex potential of period $q_1\mathbb{Z} \oplus \cdots \oplus q_d\mathbb{Z}$ Floquet isospectral to zero if and only if*

$$\max_{1 \leq j \leq d} q_j \geq 4. \tag{17}$$

We prove the period-three assertion directly. The final classification then uses the previously established existence and period-reduction results of [4].

6.1 The Floquet gauge and coordinate-average descent

Write $G_d = (\mathbb{Z}/3\mathbb{Z})^d$ and $\omega = e^{2\pi i/3}$. For $t = (t_1, \ldots, t_d) \in (\mathbb{C}^*)^d$, use the Floquet gauge $f(n) = t_1^{n_1} \cdots t_d^{n_d} \phi(n)$, where ϕ is three-periodic in each coordinate. The Floquet multipliers are t_j^3, and the gauged matrix on functions on G_d is

$$H_V(t) = \sum_{j=1}^{d} (t_j S_j + t_j^{-1} S_j^{-1}) + \operatorname{diag}(V), \qquad (S_j \phi)(n) = \phi(n + e_j). \tag{18}$$

Floquet isospectrality gives

$$\det(\lambda I - H_V(t)) = \det(\lambda I - H_0(t)) \qquad (t \in (\mathbb{C}^*)^d). \tag{19}$$

Indeed, the equality initially holds on the product unit torus. Each characteristic-coefficient difference is a Laurent polynomial in t; the one-variable polynomial identity theorem, applied successively to its variables, extends the equality to $(\mathbb{C}^*)^d$.

For a coordinate j, define the averaged potential on G_{d-1} by

$$(\mathcal{A}_j V)(n_1, \ldots, \widehat{n_j}, \ldots, n_d) = \frac{1}{3} \sum_{r=0}^{2} V(n_1, \ldots, r, \ldots, n_d).$$

Lemma 6.1 (Coordinate-average descent)**.** *If V satisfies (19), then $\mathcal{A}_j V$ is Floquet isospectral to zero on G_{d-1}.*

Proof. Set $N = 3^{d-1}$, fix all parameters except t_j, and write

$$H_V(t) = t_j S_j + t_j^{-1} S_j^{-1} + B + \operatorname{diag}(V), \qquad B = \sum_{\ell \neq j} (t_\ell S_\ell + t_\ell^{-1} S_\ell^{-1}).$$

Fourier transformation in the jth coordinate decomposes the vector space into three N-dimensional eigenspaces of S_j, with eigenvalues $1, \omega, \omega^2$. The compression of B to each of

them is the lower-dimensional free matrix B. The compression of $\operatorname{diag}(V)$ is $\operatorname{diag}(\mathcal{A}_j V)$, as follows by taking the normalized inner product in the jth coordinate.

Fix $k \in \{0,1,2\}$ and put $\lambda = \omega^k t_j + \mu$. The two other diagonal blocks of $\lambda I - H_V(t)$ have leading terms $(\omega^k - \omega^r) t_j I_N$, $r \neq k$, whereas all off-diagonal blocks are bounded as $t_j \to \infty$. Block elimination, or expansion of the determinant by its leading power of t_j, gives

$$\lim_{t_j \to \infty} t_j^{-2N} \det\bigl((\omega^k t_j + \mu) I - H_V(t)\bigr) = \prod_{r \neq k} (\omega^k - \omega^r)^N \det\bigl(\mu I - B - \operatorname{diag}(\mathcal{A}_j V)\bigr).$$

The same limit for $V = 0$ has the averaged potential absent. The nonzero constant factors cancel in (19). This proves the lower-dimensional characteristic-polynomial identity for every μ and every choice of the other parameters. □

Expand the potential in characters:

$$V(n) = \sum_{a \in G_d} c_a \omega^{a \cdot n}, \qquad c_a = 3^{-d} \sum_{n \in G_d} V(n) \omega^{-a \cdot n}. \tag{20}$$

If rigidity has been established in dimension $d-1$, Lemma 6.1 gives $\mathcal{A}_j V = 0$ for every j. In Fourier coordinates this means

$$c_a = 0 \quad \text{whenever some } a_j = 0. \tag{21}$$

The only possible remaining frequencies therefore lie in $\mathcal{S}_d = \{1, -1\}^d \subset G_d$, and in particular $c_0 = 0$. The zero-dimensional initial case is a scalar: equality of its spectrum with zero forces that scalar to vanish. This supplies the base for the dimension induction.

6.2 Exact second- and third-order identities

We now assume (21) and derive the identities that eliminate the remaining coefficients. Set $t_j = s x_j$ and $\varepsilon = s^{-1}$, where the x_j are independent nonzero variables. In the character basis, division of (18) by s gives

$$K(\varepsilon) = D + \varepsilon C + \varepsilon^2 E, \quad D = \operatorname{diag}(L_a), \quad E = \operatorname{diag}(M_a), \quad C_{ab} = c_{a-b}, \tag{22}$$

where

$$L_a = \sum_{j=1}^{d} x_j \omega^{a_j}, \qquad M_a = \sum_{j=1}^{d} x_j^{-1} \omega^{-a_j}.$$

Work over the rational function field $F = \mathbb{C}(x_1, \dots, x_d)$. The elements L_a are pairwise distinct: different indices have different coefficient vectors in their linear forms. Thus the free eigenvalue through L_0 is simple, and (19) forces the corresponding formal eigenvalue of $K(\varepsilon)$ to be exactly

$$L_0 + \varepsilon^2 M_0.$$

There is no ambiguity from a permutation of branches, since their constant terms are distinct. Define their gaps

$$P_a = L_0 - L_a = \sum_{j=1}^{d} (1 - \omega^{a_j}) x_j \qquad (a \neq 0). \tag{23}$$

Lemma 6.2 (The two perturbation identities)**.** *The Fourier coefficients satisfy the following rational identities in F:*

$$\sum_{a\neq 0} \frac{c_a c_{-a}}{P_a} = 0, \tag{24}$$

$$\sum_{a,b\neq 0} \frac{c_{-a} c_{a-b} c_b}{P_a P_b} = 0. \tag{25}$$

Proof. Choose the formal eigenvector $u(\varepsilon) = e_0 + \varepsilon u^{(1)} + \varepsilon^2 u^{(2)} + \cdots$ with zeroth coordinate equal to one. Such a vector exists uniquely by the simple-eigenvalue equation over $F[[\varepsilon]]$; in particular, $u_0^{(r)} = 0$ for every $r > 0$. The order-one equation from $K(\varepsilon)u = (L_0 + \varepsilon^2 M_0)u$ gives

$$u_a^{(1)} = \frac{C_{a0}}{P_a} \qquad (a \neq 0).$$

At order two, the nonzero coordinates give

$$u_a^{(2)} = \sum_{b\neq 0} \frac{C_{ab} C_{b0}}{P_a P_b}.$$

Here Ee_0 has no nonzero coordinate. The zeroth coordinate at the same order is $\sum_{a\neq 0} C_{0a} u_a^{(1)} + M_0 = M_0$, which is (24). At order three, the zeroth coordinate gives $\sum_{a\neq 0} C_{0a} u_a^{(2)} = 0$, yielding (25). Both $Eu^{(1)}$ and $M_0 u^{(1)}$ have zero zeroth coordinate. This accounts for all possible contributions from the $\varepsilon^2 E$ term; there is no omitted mixed term. □

By (21), the nonzero terms in the quadratic identity have $a \in \mathcal{S}_d$. In a nonzero term of the cubic identity the three steps $-a$, $a - b$, and b all lie in $\mathcal{S}_d$. In each coordinate they sum to zero in $\mathbb{Z}/3\mathbb{Z}$. Three signs in $\{1, -1\}$ sum to zero modulo three only when all three are equal. It follows that $b = -a$, and the cubic identity becomes

$$\sum_{\{a,-a\}\subset \mathcal{S}_d} \frac{c_a^3 + c_{-a}^3}{P_a P_{-a}} = 0, \tag{26}$$

where the sum takes one representative of each antipodal pair.

6.3 Separation by the gap hyperplanes

Put $A = 1 - \omega$, $B = 1 - \omega^2$, and $S = x_1 + \cdots + x_d$. Then

$$A + B = AB = 3, \qquad A \neq B, \qquad A^2 \neq B^2.$$

For a sign vector a, the coefficients of P_a are A and B. Call a mixed if both signs occur.

Lemma 6.3 (Distinct mixed gap hyperplanes)**.** *If $a \in \mathcal{S}_d$ is mixed, P_a is not proportional to P_b for any distinct $b \in \mathcal{S}_d$. The only proportional pair of gap forms is that for the two constant sign vectors, with forms AS and BS.*

Proof. Suppose $P_b = rP_a$. Since both coefficient values occur in P_a, the nonzero scalar r must map the two-element set $\{A, B\}$ to itself. If it fixes both elements then $r = 1$ and $b = a$. If it interchanges them, then $rA = B$ and $rB = A$, contradicting $A^2 \neq B^2$. A form with two different coefficient values cannot be proportional to a constant-coefficient form. The assertion for the two constant vectors is immediate. □

For mixed a, multiply (24) by P_a and reduce modulo the prime ideal (P_a) in $\mathbb{C}[x_1, \dots, x_d]$, after localizing the other denominator factors. By Lemma 6.3, only its single term survives. Hence $c_a c_{-a} = 0$ for every mixed a. Let c_+ and c_- denote the coefficients of the two constant sign vectors. The remaining quadratic identity is

$$\frac{c_+ c_-}{S}\left(\frac{1}{A} + \frac{1}{B}\right) = \frac{c_+ c_-}{S} = 0.$$

We have therefore established

$$c_a c_{-a} = 0 \qquad (a \in \mathcal{S}_d). \tag{27}$$

Apply the same operation to (26). For mixed a, its denominator factor P_a occurs in only that pair, and P_{-a} is nonzero modulo (P_a). This gives $c_a^3 + c_{-a}^3 = 0$. The remaining constant pair has denominator ABS^2, so its numerator also vanishes. Thus

$$c_a^3 + c_{-a}^3 = 0 \qquad (a \in \mathcal{S}_d). \tag{28}$$

Equations (27) and (28) imply $c_a = c_{-a} = 0$: if one were nonzero, the product equation would force the other to be zero, contradicting the cube equation. In dimension one there are no mixed vectors, and the constant-pair calculations above give the entire argument without any hyperplane restriction.

Proof of Theorem T79. We induct on the dimension, beginning with the scalar case $d = 0$. For a zero-isospectral potential in dimension d, the induction hypothesis and Lemma 6.1 give (21). Lemma 6.2 and the gap-hyperplane argument then show that every remaining Fourier coefficient is zero. Fourier inversion proves $V \equiv 0$. This establishes rigidity for every $(3\mathbb{Z})^d$.

For the rectangular classification, [4, Proposition 4.2] states the following exact equivalence: when the last period is one or two, a nonzero zero-isospectral potential exists in that dimension if and only if one exists after deleting that coordinate. Repeatedly apply this equivalence to a period vector with all $q_j \leq 3$. The remaining vector has all entries equal to three, where rigidity has just been proved, or has dimension zero. Thus no nonzero potential exists. Conversely, if some $q_j > 3$, existence is [4, Theorem C]. These two directions prove (17). □

The proof uses identities in independent Floquet variables. It does not infer rigidity from a single fixed Floquet matrix or impose reality, separability, or smaller least period on the potential. The explicitly open family in [4, Section 4.4] is settled in full by the period-three assertion. The final if-and-only-if statement completes the rectangular classification by adjoining the cited prior results. General oblique lattices remain outside this conclusion.

7 T80: optimal quadratic stabilization in equivariant Ehrhart theory

The target in this section is Conjecture 8.5 of Clarke [6]. It is a coefficientwise inequality for the equivariant Ehrhart series of an arbitrary lattice polytope with an involution. We first formulate and solve a parameter-construction problem, then apply its answer to that conjecture without a polynomiality or integer-decomposition assumption on the polytope.

7.1 The original inequality and the auxiliary problem

Let L be a lattice of rank $n + 1$, equipped with a primitive height functional. Let P be a full-dimensional lattice polytope in the affine hyperplane of height one, preserved by a

nontrivial involution σ of L. Write $G = \{1, \sigma\}$ and let ε denote its sign character. Let p and q be the dimensions of the $+1$ and -1 eigenspaces of the ambient representation. Thus

$$p + q = n + 1, \qquad p \geq 1, \qquad q \geq 1.$$

The restriction $q \geq 1$ excludes the trivial action; it is necessary for the universal optimality statement below.

Let $R = \mathbb{C}[\mathrm{cone}(P) \cap L]$ be the Ehrhart ring, graded by height. Its equivariant Hilbert series is

$$E_P(t) = \sum_{j \geq 0} [R_j] t^j = \frac{\varphi_P(t)}{D(t)}, \qquad D(t) = (1-t)^p (1 - \varepsilon t)^q,$$

in the representation ring $R(G) = \mathbb{Z}[\varepsilon]/(\varepsilon^2 - 1)$. An element of this ring is *effective* if it is the character of an actual representation, equivalently if both its trivial and sign multiplicities are nonnegative. The same term is used coefficientwise for a power series.

The evaluation $\varphi_P(t)(1)$ is the ordinary polynomial $h^*_P(t)$. The fixed polytope has denominator dividing two. Write its Ehrhart series as $H(t)/(1-t^2)^p$, where H has nonnegative integer coefficients. Then

$$\varphi_P(t)(\sigma) = (1+t)^{q-p} H(t). \tag{29}$$

These are the conventions of [6, Section 2 and Lemma 8.1]. Clarke's Conjecture 8.5 asserts

$$\begin{cases} (1+t)^{p-q} h^*_P(t) \;\geq\; H(t), & p \geq q, \\ h^*_P(t) \;\geq\; (1+t)^{q-p} H(t), & p < q, \end{cases} \tag{30}$$

where each inequality compares coefficients.

Our auxiliary question is the following: how many invariant quadratic parameters must replace invariant linear parameters to obtain a homogeneous system of parameters of R, while keeping exactly q anti-invariant linear parameters? A second question asks for the least exponent $s = s(p, q)$ such that $(1+t)^s \varphi_P(t)$ is effective for *every* polytope of ambient type (p, q). The first answer depends on a base locus; the universal answer is $\max(p - q, 0)$.

Put $X = \operatorname{Proj} R$, so $\dim X = p + q - 1$. Write W_+ and W_- for the invariant and anti-invariant subspaces of R_1, and use the base-locus convention

$$Z_+ = V(W_-), \qquad Z_- = V(W_+), \qquad b = \dim Z_-.$$

If Z_- is empty, set $b = -\infty$ and interpret $\max(0, b - q + 1)$ as zero.

7.2 Base loci and generic sections

Lemma 7.1. *The space R_1 is basepoint-free on X, and*

$$Z_+ \cap Z_- = \varnothing, \qquad \dim Z_+ = p - 1, \qquad \dim Z_- \leq p - 1.$$

The invariant subspace $(R_2)^G$ is also basepoint-free.

Proof. The linear base-locus assertions are [6, Lemmas 7.1(a) and 7.3]. We recall their geometric content to make clear that no assumption that R is generated in degree one is being made. Every point of the projective toric variety X lies on the orbit associated to a nonempty face F of P. A monomial corresponding to any vertex of F is nonzero at that point. Hence R_1 has no base point, and the two eigenspace base loci are disjoint.

On a face orbit, the equations defining either Z_+ or Z_- force the face to be σ-invariant: otherwise a vertex monomial supported on F is paired with a monomial vanishing on that orbit. For an invariant face, the equations on its torus characters have the form $\chi(\sigma u) = \chi(u)$ or $\chi(\sigma u) = -\chi(u)$ for its lattice points u. Their solution set is either empty or a coset of a torus whose dimension, after quotienting by projective scaling, is $\dim F^\sigma \le p-1$. For the invariant equations on the full face P, the trivial character supplies a solution and the dimension is $p-1$. Taking the finite union over face orbits gives the asserted bounds.

Finally, at any point of X, at least one eigenlinear form $f \in W_+ \cup W_-$ is nonzero. Its square f^2 is an invariant element of degree two, still nonzero there. This proves the last assertion. □

Lemma 7.2. *Let $0 \le k \le p$. A homogeneous system of parameters consisting of $p-k$ invariant linear forms, q anti-invariant linear forms, and k invariant quadratic forms exists if and only if*

$$k \ge \delta(P) := \max(0, b - q + 1). \tag{31}$$

Proof. First suppose (31) holds. Choose the linear forms from the affine parameter space

$$\mathcal{U} = W_+^{p-k} \times W_-^{q}.$$

Over $X \setminus (Z_+ \cup Z_-)$, the condition that all chosen forms vanish at a specified point imposes $p-k+q$ independent linear conditions on $\mathcal{U}$: each evaluation acts nontrivially in its own tuple coordinate. Over Z_+, it imposes $p-k$ conditions; over Z_-, it imposes q. Thus the corresponding incidence strata have dimensions at most, respectively,

$$\begin{aligned} \dim \mathcal{U} + (p+q-1) - (p-k+q) &= \dim \mathcal{U} + k - 1, \\ \dim \mathcal{U} + (p-1) - (p-k) &= \dim \mathcal{U} + k - 1, \\ \dim \mathcal{U} + b - q &\le \dim \mathcal{U} + k - 1. \end{aligned}$$

The generic-fiber dimension theorem now implies that a general tuple of linear forms has common zero locus $Y \subset X$ of dimension at most $k-1$. A stratum not dominating $\mathcal{U}$ has empty generic fiber and does not affect this conclusion. In particular, if $k = 0$, then Y is empty.

By Lemma 7.1, invariant quadratics are basepoint-free. Therefore k general invariant quadratic sections have no common zero on Y: successively they cut positive-dimensional components by one dimension and avoid any remaining finite set of points. The full collection has no common projective zero. By the graded Nullstellensatz, its homogeneous ideal has radical equal to the irrelevant ideal, and the quotient ring is finite-dimensional. Since there are $p+q = \dim R$ forms, they are a homogeneous system of parameters.

Conversely, consider any system with the stated degree and character pattern, and suppose Z_- is nonempty. On a component of Z_- of dimension b, all invariant linear forms vanish identically. The remaining $q+k$ positive-degree sections must therefore have no common zero on that projective component. This requires $q+k \ge b+1$: at most b positive-degree hypersurfaces cannot have empty intersection on a projective variety of dimension b. Equivalently, its homogeneous coordinate ring has dimension $b+1$, and Krull's height theorem prevents an ideal generated by at most b positive-degree elements from having a finite quotient. Hence $k \ge b - q + 1$. Together with $k \ge 0$, this proves necessity, including the empty-base-locus case. □

7.3 The stabilization theorem and its application

Theorem T80 (Optimal quadratic stabilization and Clarke's inequality)**.** *For the order-two lattice action above, set*

$$\delta(P) = \max(0, \dim V(W_+) - q + 1), \qquad r = \max(p - q, 0).$$

Then $0 \le \delta(P) \le r$, and there is a finite-dimensional graded G-algebra with character

$$(1+t)^{\delta(P)}\varphi_P(t). \tag{32}$$

The number $\delta(P)$ is the minimum number of invariant quadratics in the parameter pattern of Lemma 7.2.

For each fixed pair $p, q \ge 1$, the least integer $s \ge 0$ for which $(1+t)^s\varphi_P(t)$ is effective for every polytope of ambient type (p, q) is exactly r. In particular, the original coefficient inequalities (30), namely Conjecture 8.5 of [6], hold for every such polytope.

Proof. The bound on $\delta(P)$ follows from Lemma 7.1. Apply Lemma 7.2 with $k = \delta(P)$. The Ehrhart ring is a normal affine semigroup ring and is Cohen–Macaulay by Hochster's theorem [8]. Consequently its homogeneous system of parameters is a regular sequence. Taking its equivariant Koszul complex gives the character of the Artinian quotient as

$$E_P(t)(1-t)^{p-\delta(P)}(1-\varepsilon t)^q(1-t^2)^{\delta(P)} = \varphi_P(t)(1+t)^{\delta(P)}.$$

This is an actual graded representation, proving (32). The parameter minimality is precisely the necessity assertion of Lemma 7.2.

Since $r \ge \delta(P)$, multiplication by the trivially acted-on polynomial $(1+t)^{r-\delta(P)}$ preserves effectiveness. Thus $(1+t)^r\varphi_P(t)$ is always effective. Its evaluations at the identity and at σ are

$$(1+t)^r h_P^*(t), \qquad (1+t)^{r+q-p}H(t),$$

respectively. For an effective character of a group of order two, its identity value is at least the absolute value of its value at σ. The second displayed polynomial has nonnegative coefficients. Comparing each degree therefore proves exactly (30) in both cases $p \ge q$ and $p < q$.

It remains to prove the sharpness of the universal exponent. There is nothing to prove when $p \le q$, since then $r = 0$. Suppose $r = p - q > 0$. Start with the cube $C = [0,1]^{r+1}$ and reflect its first coordinate:

$$(x_1, x_2, \dots, x_{r+1}) \longmapsto (1 - x_1, x_2, \dots, x_{r+1}).$$

On the homogeneous lattice $L_0 = \mathbb{Z}^{r+2}$ with coordinates $(x_1, \dots, x_{r+1}, h)$, this is the integral involution

$$(x_1, x_2, \dots, x_{r+1}, h) \longmapsto (h - x_1, x_2, \dots, x_{r+1}, h).$$

Its ambient type is $(r+1, 1)$. The fixed polytope has $x_1 = 1/2$, so its lattice-point count in dilation j is $(j+1)^r$ for even j, and zero for odd j. Its Ehrhart series is therefore

$$E_{C^\sigma}(t) = \sum_{\ell \ge 0} (2\ell+1)^r t^{2\ell}.$$

The generating series of the degree-r polynomial $(2\ell+1)^r$ has a genuine pole of order $r+1$ at $t^2 = 1$: its leading contribution is $2^r r!/(1-t^2)^{r+1}$. In particular, the pole at $t = -1$ has order $r+1$. Multiplication by $D_C(t)(\sigma) = (1-t)^{r+1}(1+t)$ leaves a pole of order exactly r in $\varphi_C(t)(\sigma)$.

To obtain arbitrary $q \ge 1$ while retaining this pole, take the standard lattice join with $q-1$ unimodular segments, each with the involution swapping its endpoints. More explicitly, use

$$L = L_0 \oplus (\mathbb{Z}^2)^{q-1}, \qquad \mathrm{ht}(x, h; y) = h + \sum_{a=1}^{q-1} (y_{a,0} + y_{a,1}),$$

and take the convex hull of the embedded $(C, 1)$ and the embedded segments $\mathrm{conv}(e_{a,0}, e_{a,1})$ in height one. Apply the cube involution on L_0 and swap each added coordinate pair. The cone

over this join is the product of the cube cone with $q-1$ nonnegative quadrants. Its lattice semigroup is consequently a direct product, and its Ehrhart ring is the graded tensor product of R_C with $q-1$ two-variable polynomial rings having swapped variables.

Each variable pair contributes ambient representation $1+\varepsilon$ and Hilbert series $1/((1-t)(1-\varepsilon t))$. Its determinant factor cancels, so the join has equivariant numerator exactly $\varphi_C(t)$. Its ambient type is $(r+q, q) = (p, q)$. Therefore a multiplier $(1+t)^s$ with $s < r$ leaves a pole in its evaluation at σ.

An effective series whose identity evaluation is a polynomial must itself be a polynomial: in every degree beyond the degree of that evaluation, an effective representation of dimension zero is zero. The identity evaluation of $(1+t)^s\varphi_P(t)$ is the polynomial $(1+t)^s h_P^*(t)$. Hence the remaining pole precludes effectiveness for $s < r$. This proves the universal lower bound and completes the theorem. □

The two optimality assertions must be distinguished. For a particular polytope, $\delta(P)$ is optimal for the specified degree and character pattern of parameters. It need not be the least exponent making its series effective: [6, Example 5.3] already exhibits an effective series with an obstruction to linear parameters. The exponent r is instead optimal after quantifying over all polytopes of fixed ambient type. The construction above proves that universal statement.

The base-locus description, generic incidence argument, and Koszul character formula are prior tools. The contribution proposed here is the exact mixed-degree parameter budget, its sharp universal stabilization consequence, and the resulting proof of the original coefficient inequality. This distinction concerns historical attribution; it does not add an unproved hypothesis to the mathematical argument.

8 Attribution, novelty, and the requested success conditions

The prior problem statements are pinned to specific primary sources: [1, Conjecture 4], [2, Conjecture 12], [4, Section 4.4], and [6, Conjecture 8.5]. The first two sources originate in 2015 and 2013; the latter two in 2026. The spectral target is explicitly open in the cited source, but is not numbered or given a standalone conjecture name there. It must not be relabelled as a previously named conjecture. The nested binary special case [3, Conjecture 4] is not counted as another independent target.

The arithmetic construction uses the classical congruence $B(z)^p \equiv B(z^p) \pmod p$. Its specific contribution here is the substitution-compatible correction term, which produces the required valuation at every level and a full kernel description. The Cantor proof returns every coefficient to the prescribed ring before claiming the original conjecture. No coefficient-ring enlargement is hidden in the final statement.

The spectral argument uses earlier Fourier and rational-pole techniques [5]. Its additional period-three step is that all surviving three-step Fourier cycles follow a single direction. The resulting cubic relations remove the remaining complex coefficients without a positivity assumption. The contribution is this specific closure argument, not a claim to have invented perturbation theory or pole extraction.

The Ehrhart construction extends Clarke's base-locus and generic-section framework [6]. The exact quadratic budget and its character factor are the additional ingredients. The Cohen–Macaulay and equivariant Koszul facts are prior [8, 7]. A different proof of Clarke's dilation Conjecture 8.7 was also examined, but its central determinant-ratio and coefficient-extraction mechanism is already explicit in [7, Section 2.5, equation (7), and Lemma 2.33]. It is not included as an additional new-method success in this manuscript.

Correction to the preceding T71–T75 attribution. The even-period Fourier-square quadratic structure used in the preceding T71 draft is already explicit in Korotyaev and Kutsenko [9, Theorem 2.4, Lemma 2.6, and equation (2.31)]. That structure must be credited to their 2005 preprint and subsequent publication. The correction narrows the earlier novelty description; it does not itself contradict the algebraic consequences proved in that draft. The present T79 argument is a separate multidimensional rigidity argument, and no part of T76–T80 counts that older even-period structure as a new principle.

What has and has not been established. The displayed arguments address the stated source problems unchanged and without conjectural assumptions. Searches through 24 October 2026 examined the current source versions, relevant author publication lists, and topic and conjecture-number searches using two search services. No identical earlier resolution was located. Such a search cannot prove universal absence from the literature. Nor does a new construction using classical tools establish that its general principle is historically unprecedented. Consequently this manuscript does *not* certify all six requested historical success conditions. It supplies the complete mathematical arguments and an auditable, bounded priority assessment. The numbering T76–T80 organizes five theorems, with three numbered prior conjectures and one prior unnumbered problem as applications.

A Exact checks and reproducibility

The accompanying verification bundle contains executable Python scripts and their output. All checks use exact integer or rational arithmetic. They support the proofs but establish only the finite cases actually executed.

Check	Executed range and result
Additive Frobenius tower	First 256 coefficients; $p = 2$ through level 4, $p = 3$ through level 3, and $p = 5$ through level 2. Every predicted coefficientwise divisibility holds.
Cantor tower	First 256 coefficients at levels 0, 1, 2, with degrees $2, 6, 18$ and valuations at least $1, 4, 13$. The displayed original-ring degree-six relation modulo 81 also passes.
Spectral pole separation	Dimensions 1 through 5: the evaluated rational bases have exact ranks $1, 2, 4, 8, 16$, and every admissible three-step cycle uses a single Fourier direction.
Reflection cubes	Dimensions 2 through 7: exact Ehrhart and fixed-point numerators give nonnegative integral character multiplicities after the predicted stabilization; the pole order at -1 is the claimed one.

The full proofs were also checked by independent internal derivations: coefficient isolation and valuation descent for T76–T78; formal eigenvector recursion and hyperplane residues for T79; and incidence dimensions, non-degree-one generation, and equivariant Koszul factors for T80. These checks should not be described as external peer review or formal machine verification.

The source is standalone: compiling it twice with `pdflatex` requires no separate figures or bibliography files. The verification bundle includes its own run instructions and records the exact finite scope of each script.

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